LABSHEET-11

Q1. Create a directory named archive_lab and create three text files inside it. Add sample content to each file.

```
mkdir -p archive_lab
echo "File 1 content" > archive_lab/file1.txt
echo "File 2 content" > archive_lab/file2.txt
echo "File 3 content" > archive_lab/file3.txt
```

Q2. Create a tar archive named files.tar containing all files from archive_lab. Verify that the archive was created.

```
tar -cvf files.tar -C archive_lab .
ls -l files.tar
```

Q3. List the contents of files.tar without extracting it.

```
tar -tvf files.tar
```

Q4. Extract files.tar into a new directory named extracted_tar.

```
mkdir -p extracted_tar
tar -xvf files.tar -C extracted_tar
ls -l extracted_tar
```

Q5. Create a gzip-compressed file from a text file and observe the change in its filename.

```
gzip archive_lab/file1.txt
ls -l archive_lab
```

Q6. Use gunzip to decompress the gzip-compressed file and verify that the original file is restored.

```
gunzip archive_lab/file1.txt.gz
ls -l archive_lab
```

Q7. Create a tar.gz archive named archive.tar.gz containing the complete archive_lab directory.

```
tar -czvf archive.tar.gz archive_lab
ls -l archive.tar.gz
```

Q8. List the contents of archive.tar.gz without extracting it.

```
tar -tzvf archive.tar.gz
```

Q9. Extract archive.tar.gz into a separate directory and verify the extracted files.

```
mkdir -p extracted_targz
tar -xzvf archive.tar.gz -C extracted_targz
ls -l extracted_targz/archive_lab
```

Q10. Create a zip archive named files.zip containing three selected files.

```
zip files.zip archive_lab/file1.txt archive_lab/file2.txt archive_lab/file3.txt
ls -l files.zip
```

Q11. List the contents of files.zip without extracting it.

```
unzip -l files.zip
```

Q12. Extract files.zip into a directory named extracted_zip.

```
unzip files.zip -d extracted_zip
ls -l extracted_zip
```

Q13. Use unzip to extract only one selected file from a ZIP archive.

```
unzip files.zip archive_lab/file1.txt -d single_extracted
```

Q14. Create a ZIP archive containing an entire directory and verify its contents.

```
zip -r archive_dir.zip archive_lab
unzip -l archive_dir.zip
```

Q15. Create a gzip-compressed tar archive and explain the difference between tar and gzip in the command sequence used.

```
# tar creates an uncompressed archive of files, while gzip compresses a single file/archive.
# Combined command using -z flag:
tar -czvf archive.tar.gz archive_lab
```

Q16. Create a tar archive, add another file to the archive, and verify that the new file is present.

```
tar -cvf test.tar archive_lab/file1.txt
echo "File 4 content" > file4.txt
tar -rvf test.tar file4.txt
tar -tvf test.tar
```

Q17. Compare the archive contents and file sizes of a tar archive and a compressed tar.gz archive.

```
tar -cvf sample.tar archive_lab
tar -czvf sample.tar.gz archive_lab
ls -lh sample.tar sample.tar.gz
```

Q18. Use appropriate commands to decompress and extract a tar.gz archive and a ZIP archive into separate directories.

```
mkdir -p dir_targz dir_zip
tar -xzvf archive.tar.gz -C dir_targz
unzip files.zip -d dir_zip
```

Q19. Create a complete backup archive of a practice directory using tar.gz, then restore it into another directory.

```
# Create backup:
tar -czvf backup.tar.gz archive_lab


# Restore backup:
mkdir -p restored_backup
tar -xzvf backup.tar.gz -C restored_backup
```

Q20. Demonstrate a complete archive and compression workflow using tar, gzip, gunzip, zip, and unzip, including verification after each operation.

```
# 1. Create tar & gzip:
tar -czvf demo.tar.gz archive_lab
```

```
tar -tzvf demo.tar.gz

# 2. Extract tar.gz:
mkdir -p demo_tar_out
tar -xzvf demo.tar.gz -C demo_tar_out

# 3. Create & Extract zip:
zip -r demo.zip archive_lab
unzip -l demo.zip
mkdir -p demo_zip_out
unzip demo.zip -d demo_zip_out
```